

30-Day SDET Technical Interview Preparation Plan - [Naveen Automation Labs]

Week 1: Programming & Data Structures

Days 1-3: Core Programming

- Data Structures: Arrays, Lists, HashMaps, Stacks, Queues
- Time/Space complexity basics
- Solve 3 LeetCode easy problems daily
- OOP concepts review

Days 4-7: Advanced Programming

- Trees, Graphs basics (Optional)
- Sorting and searching algorithms
- String manipulation
- Design patterns (Page Object, Factory, Builder, Strategy etc..)

Week 2: Test Automation Fundamentals

Days 8-10: Framework Basics

- TestNG/JUnit essentials
- Test architecture principles
- Assertions and annotations
- Build one basic test framework

Days 11-14: Web Automation

- Selenium WebDriver core concepts
- Locators and waits
- Actions class
- Page Object implementation
- Build small e-commerce test project

Week 3: API & Advanced Testing

Days 15-17: API Testing

- REST principles
- RestAssured/Postman/Newman
- JSON parsing
- Authentication - OAuth1, OAuth2, Basic Auth, API Key, JWT token
- Create API test framework
- HTTP methods and status codes
- Mocking - Postman mocking server, Wiremock

Days 18-21: Advanced Topics

- CI/CD basics (Jenkins/GitHub Actions)
- Docker fundamentals - Docker commands, Images, Dockerfile, Build docker images, Image mounting, Start containers
- Selenium Grid concepts - Selenium Grid 4.x, Selenoid
- Cross-browser testing
- Test reporting (Extent/Allure)

Week 4: Interview Preparation

Days 22-24: System Design

- Test architecture patterns
- Framework selection criteria
- Configuration management
- Design one end-to-end framework

Days 25-27: Mock Interviews

- Technical coding practice
- System design exercises
- Testing scenarios
- 1 mock interviews daily

Days 28-30: Final Preparation

- STAR method practice
- Project explanations
- Common behavioral questions
- Framework optimization
- Final mock interviews

Daily Schedule

1. Morning (2 hours)

- Coding practice
- Algorithm study

2. Afternoon (3 hours)

- Framework learning
- Project building

3. Evening (1 hour)

- Interview preparation
- Documentation review

Key Projects to Complete

1. Test automation framework with Selenium
2. REST API testing framework
3. CI/CD pipeline with automated tests

Resources

- LeetCode
- Naveen Automation Labs
- Online training/courses
- GitHub sample projects

Success Metrics

- 50 solved coding problems
- 3 completed projects
- 4 mock interviews
- 1 end-to-end framework